



- 3 (a) (i) State what is meant by the Avogadro constant.

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..... [1]

- (ii) State the relationship between the Avogadro constant N_A , the molar gas constant R and the Boltzmann constant k .

[1]

- (b) Two samples X and Y of ideal gases are both at thermodynamic temperature T .

Sample X has volume V and consists of N molecules, each of mass m .

Sample Y has volume $2V$ and consists of $2N$ molecules, each of mass $2m$.

- (i) Complete Table 3.1 by giving expressions, in terms of some or all of N , m , T , V and the constants in (a)(ii), for the quantities indicated.

Table 3.1

	sample X	sample Y
pressure		
amount of substance		
mean-square speed of molecules		
internal energy		

[4]





(ii) The temperature of sample X is now varied.

On Fig. 3.1, sketch the variation with thermodynamic temperature of the root-mean-square (r.m.s.) speed of the molecules of the gas.

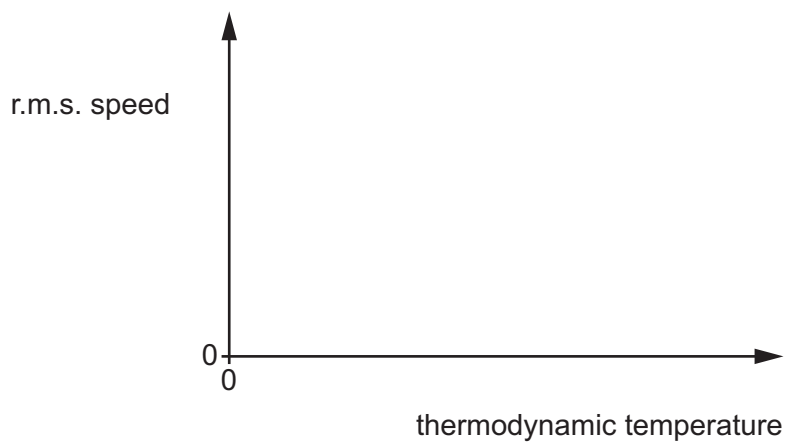


Fig. 3.1

[2]

[Total: 8]

